# NextSource:
# An Adaptive Next-Source Selection Algorithm for OSINT Agents

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# ABSTRACT

Open Source Intelligence, or OSINT, is used to collect and study information from publicly available sources such as official websites, news platforms, public databases, and technical repositories. AI agents can support this process by searching, collecting, and analyzing information. However, they may repeat similar searches, choose unsuitable source types, or stop before enough evidence has been collected.

This research proposes **NextSource**, an adaptive next-source selection algorithm for OSINT agents. The system first studies the investigation question and checks which information is already available and which information is still missing. It then ranks a predefined set of source categories, such as official sources, news sources, public databases, and technical sources. The ranking is based on factors such as expected usefulness, source reliability, possibility of duplicate information, previous search results, and search cost.

After the selected source is searched, the system updates the investigation state and decides whether another source should be searched, whether the source category should be changed, or whether the collection process should stop. The proposed method will be compared with random selection, fixed order selection, relevance based selection, gap based selection, and direct LLM based selection.

The expected result is that **NextSource** will help OSINT agents collect more useful evidence with fewer, more focused, and less repetitive searches.

**Keywords**: OSINT, AI Agent, Source Selection, Evidence Collection, Adaptive Retrieval.

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# CHAPTER 1

# INTRODUCTION

Open-Source Intelligence, commonly known as OSINT, is the process of collecting and analyzing information that is publicly available. This information may come from government websites, news platforms, public databases, research publications and technical repositories. OSINT is used in areas such as cybersecurity, defense, journalism and crisis analysis.

An OSINT investigation may require information from many different sources. Some sources may provide repeated, incomplete, outdated or conflicting information. Manually searching and comparing these sources can take considerable time. Artificial intelligence can support this process by understanding an investigation question, generating search queries, using external tools and analyzing retrieved documents.

However, using an AI agent does not automatically make the collection process effective. The agent may repeatedly search similar websites, retrieve duplicate information or overlook a source category that could provide the missing evidence. A systematic review of AI based OSINT research found that most existing studies focus on processing and analyzing collected information, while the planning stage remains less explored [1].

Recent studies have improved different parts of information retrieval. DRAGIN identifies when additional information is needed and what should be retrieved [2]. MBA-RAG selects retrieval strategies while considering question complexity and cost [3]. InfoGain-RAG measures the usefulness of retrieved documents [4], while RA-RAG considers the reliability of information sources [5]. However, these methods mainly focus on queries, retrieval strategies or documents. They do not directly decide which OSINT source category should be searched next.

This project proposes NextSource, an adaptive next-source selection algorithm for OSINT agents. The system identifies what information has already been collected and what is still missing. It then ranks predefined source categories, such as official sources, news sources, public databases and technical repositories, based on their expected usefulness, reliability, duplication risk and search cost.

After each search, the investigation state is updated and the source categories are ranked again. The purpose of NextSource is to make OSINT collection more focused, efficient and explainable by helping the agent decide **where to search next and why**.

# CHAPTER 2

# LITERATURE SURVEY

**Table 2.1: Literature Survey 1**

| | |
|---|---|
| Name of paper | A Systematic Review on Research Utilising Artificial Intelligence for Open Source Intelligence (OSINT) Applications |
| Authors | Thomas Oakley Browne, Mohammad Abedin and Mohammad Jabed Morshed Chowdhury |
| Publication Details | International Journal of Information Security, Vol. 23, pp. 2911-2938, Springer Nature, June 2024 |
| Research Findings | • The study systematically reviewed 163 research papers that applied artificial intelligence or machine learning techniques to OSINT related tasks.<br>• AI-based OSINT research has been applied across several security-related areas, including cybersecurity, law enforcement, intelligence analysis, misinformation detection, and threat intelligence. |
| Research Gap | • Limited research has addressed the planning, evaluation, and dissemination stages of the intelligence cycle compared with collection, processing, and analysis.<br>• Only a small number of studies integrated AI models with existing OSINT tools, and relatively few studies developed complete AI enabled OSINT systems suitable for practical use. |
| Relevance with Our Work | • It supports the development of an AI assisted OSINT framework that can work with different source types and maintain an explainable collection process rather than functioning only as a classification or detection model. |

**Table 2.2: Literature Survey 2**

| | |
|---|---|
| Name of paper | DRAGIN: Dynamic Retrieval Augmented Generation based on the Real-time Information Needs of Large Language Models |
| Authors | Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu and Yiqun Liu |
| Publication Details | Proceedings of the 62nd Annual Meeting of the ACL, pp. 12991-13013, August 2024 |
| Research Findings | • DRAGIN is a dynamic retrieval framework that decides when an AI model needs external information and what information should be searched.<br>• It uses uncertainty and important words from the full context to create better search queries. Tests on four question-answering datasets showed that it performed better than the compared retrieval methods. |
| Research Gap | • DRAGIN requires access to internal self-attention values of the language model, so it may not work directly with closed APIs that do not provide this information.<br>• The system was mainly tested on prepared Wikipedia-based datasets. Its performance on live websites, changing information, and specialized real-world tasks was not studied. |
| Relevance with Our Work | • DRAGIN shows that retrieval can change according to the current information need instead of following a fixed search process.<br>• NextSource uses this adaptive idea but focuses on selecting which OSINT source category, such as official sources, news, or databases, should be searched next. |

**Table 2.3: Literature Survey 3**

| | |
|---|---|
| Name of paper | MBA-RAG: a Bandit Approach for Adaptive Retrieval-Augmented Generation through Question Complexity |
| Authors | Xiaqiang Tang, Qiang Gao, Jian Li, Nan Du, Qi Li and Sihong Xie |
| Publication Details | Proceedings of the 31st International Conference on Computational Linguistics, pp. 3248-3254, January 2025 |
| Research Findings | • MBA-RAG uses a **multi-armed bandit algorithm** to adaptively choose the best retrieval strategy based on the complexity of a question instead of using a fixed retrieval method.<br>• It improves efficiency by balancing **answer quality and retrieval cost**, rewarding accurate answers while penalizing unnecessary retrieval steps. Experiments showed competitive performance with lower retrieval costs. |
| Research Gap | • MBA-RAG was tested only on benchmark question-answering datasets with fixed document collections, so its performance in live-web environments is still unknown.<br>• The reward function requires different settings for different tasks (single-hop and multi-hop), which means it may need manual tuning when applied to new datasets or applications. |
| Relevance with Our Work | • MBA-RAG shows that retrieval selection should be adaptive, choosing different retrieval methods based on the query instead of using a fixed approach.<br>• This idea supports NextSource, where the system will adaptively choose the most suitable OSINT source category (such as official sources, news, public databases, or technical sources) while considering both information quality and retrieval cost. |

**Table 2.4: Literature Survey 4**

| | |
|---|---|
| Name of paper | InfoGain-RAG: Boosting Retrieval-Augmented Generation through Document Information Gain-based Reranking and Filtering |
| Authors | Zihan Wang, Zihan Liang, Zhou Shao, Yufei Ma, Huangyu Dai, Ben Chen, Lingtao Mao, Chenyi Lei, Yuqing Ding and Han Li |
| Publication Details | Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pp. 7190-7204, November 2025 |
| Research Findings | • Information Gain RAG retrieves documents that provide the most new and useful information, instead of only selecting the most relevant documents.<br>• This approach improves answer quality and reduces redundant retrieval by prioritizing information that adds value to the reasoning process. |
| Research Gap | • Calculating information gain increases computational cost and may slow down retrieval.<br>• Most evaluations are performed on benchmark datasets, with limited validation in dynamic real-world or live-web environments. |
| Relevance with Our Work | • The idea of selecting information based on its usefulness aligns with NextSource's goal of choosing the most valuable OSINT sources rather than searching every source.<br>• Information gain can help NextSource reduce unnecessary searches, making source selection more efficient while maintaining answer quality. |

**Table 2.5: Literature Survey 5**

| | |
|---|---|
| Name of paper | Retrieval-Augmented Generation with Estimation of Source Reliability |
| Authors | Jeongyeon Hwang, Junyoung Park, Hyejin Park, Dongwoo Kim, Sangdon Park and Jungseul Ok |
| Publication Details | Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pp. 34279-34303, November 2025 |
| Research Findings | • RA-RAG estimates the reliability of different sources by cross-checking information across multiple sources, instead of relying only on document relevance.<br>• It retrieves information from reliable and relevant sources, then combines the evidence using weighted majority voting. Experiments showed better accuracy and scalability when sources had different reliability levels. |
| Research Gap | • The method depends on having enough overlapping information across sources to estimate reliability. It may be less effective when sources provide unique or limited information.<br>• Source reliability can change over time, but the paper does not fully address continuous reliability updates in dynamic live-web environments. |
| Relevance with Our Work | • RA-RAG supports NextSource by showing that source reliability should be considered when selecting information, not just source relevance.<br>• NextSource can use a similar idea to prioritize trustworthy OSINT categories, such as official sources and verified public databases, while reducing dependence on unreliable sources. |

# CHAPTER 3

# PROBLEM DEFINITION

1. AI-assisted OSINT systems often follow fixed or general search processes. They do not maintain a structured investigation state showing which evidence requirements are covered, partially covered, missing, or conflicting.
2. The next source is often selected using only semantic relevance or direct language model judgement. This approach does not jointly consider expected information contribution, source reliability, novelty, temporal relevance, duplication risk, and collection cost.
3. Without an adaptive collection mechanism, an OSINT agent may repeatedly retrieve similar or low-value information. It may also stop before important evidence is collected or continue searching after sufficient evidence has already been obtained.
4. Existing retrieval methods mainly focus on query generation, retrieval timing, strategy selection, or document ranking. They do not provide an explainable and well evaluated method for selecting the next OSINT source category according to the current investigation state.

# CHAPTER 4

# OBJECTIVES

1. To design a structured investigation state model that records the collected evidence and classifies each evidence requirement as covered, partially covered, missing, or conflicting.
2. To develop and implement the NextSource algorithm for ranking predefined source categories using expected information contribution, reliability, novelty, temporal relevance, duplication risk, and collection cost.
3. To develop an adaptive decision mechanism that updates the investigation state after every search and decides whether the agent should continue, change the source category, or stop the collection process.
4. To evaluate the proposed method using reproducible historical OSINT scenarios and compare it with random, fixed order, relevance-based, gap-based, and direct language model selection methods based on evidence coverage, search efficiency, duplication rate, and explainability.

# CHAPTER 5

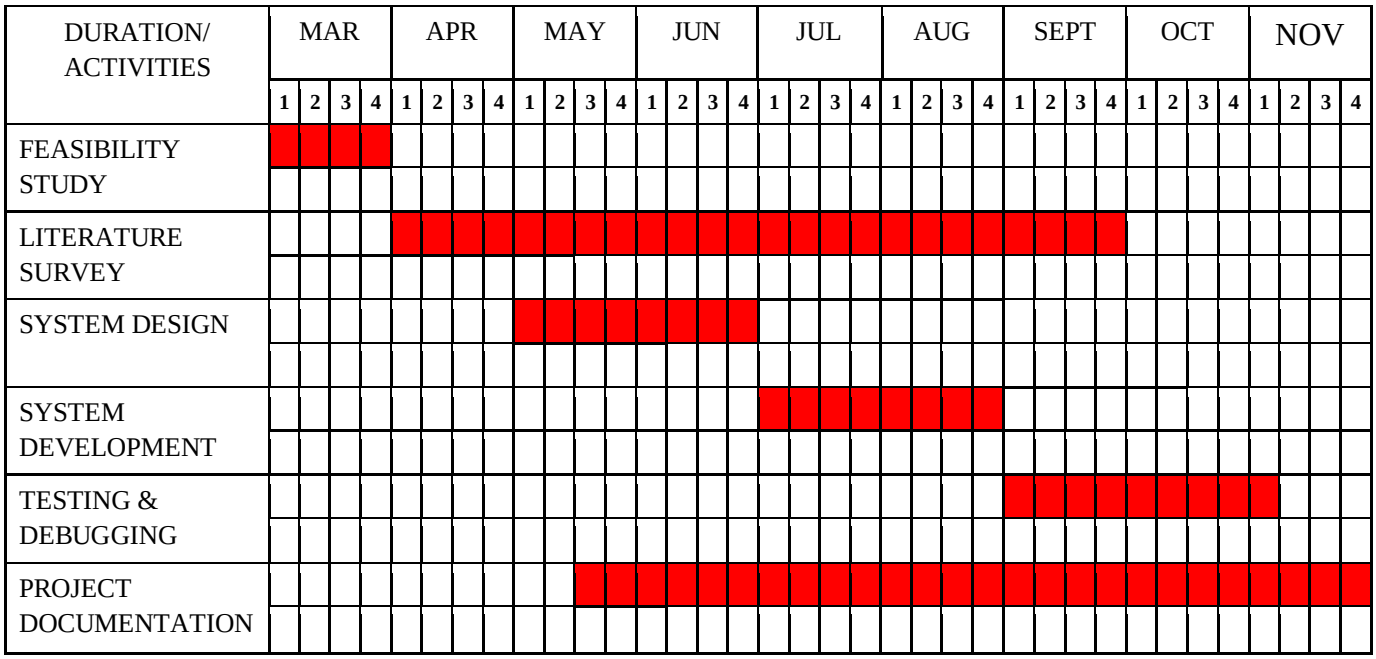
# APPLICATIONS

1. **Defense and Security Intelligence**: NextSource can help analysts identify whether the next search should focus on official sources, news reports, public databases, or technical sources while studying security-related events.
2. **Cyber Threat Intelligence**: The system can guide an AI agent toward the most useful source category when collecting information about cyber incidents, threat actors, vulnerabilities, or malware campaigns.
3. **Journalism and Fact-Checking**: NextSource can reduce repeated searches by directing researchers toward source categories that can provide missing evidence or official confirmation.
4. **Crisis and Event Monitoring**: During disasters, conflicts, or rapidly changing events, the system can help organize evidence collection and select source categories that are more likely to provide timely and useful information.
5. **Academic and Policy Research**: Researchers can use the method to collect evidence from different source categories in a more structured, efficient, and explainable manner.

# CHAPTER 6

# PLANNING

**Figure 6.1: Gantt chart.**




| | |
|---|---|
| PROPOSED ACTIVITY | |
| ACHIVE ACTIVITY | |

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